10–11 第2篇

阅读理解（说明文） The Animals That Change Colour with the Seasons

① In northern Canada, there is a small hare. In April, it is a rich chocolate brown, blending perfectly with the wet wood of the forest floor. In December, the same animal is almost **pure white**, invisible against a field of snow. No one has painted it. No human has touched it. In fact, nothing outside the hare has changed at all — the hare has changed itself. Every autumn, many animals in the northern half of our planet do the same: they quietly swap their summer coat for a snow-coloured one. **How, and why, does this happen?**

② The most famous example is the **mountain hare**. In summer the hare's fur is grey-brown. By January it is a snowy white, except for the tips of its ears, which stay black. Similar changes happen with the **Arctic fox**, the **ermine** (a small weasel), and a beautiful white-feathered bird called the **rock ptarmigan**, which changes colour three times a year to match spring grass, summer rock and winter snow.

③ The reason is **camouflage** — the art of hiding in plain sight. On a bright sandy beach, a grey pebble is easy to spot; on a snowy field, a white animal nearly disappears. Predators that live on snow — such as the snowy owl or the polar bear — cannot see a white hare against white snow until they are very close. For hunters like the Arctic fox, the same rule works in reverse: a white fox is almost invisible to the small rodents it hunts. In nature, colour is survival.

④ But how does the animal know it is time to change? You might guess that the animal "feels the cold". Surprisingly, this is only partly true. The more important signal is **day length**, not temperature. As autumn shortens the daylight hours, a small pea-sized part of the animal's brain, called the *pineal gland*, produces more of a chemical named **melatonin**. Melatonin travels through the blood and tells the tiny cells that make fur pigment to **stop producing brown colour**. Slowly, new hairs grow in without their usual dark colouring — and they appear white, because hollow fur reflects every colour of light equally.

⑤ This link with daylight, not with cold, has an interesting side-effect. In some places, winters are becoming warmer. Snow now comes later and melts earlier. But the hares still receive the old "change now" signal from their day-length clock. The result? A **white hare** standing on a brown autumn field with no snow in sight — easy for every fox and eagle to see. Scientists have called these unlucky animals "**mismatched hares**". Their survival rates in years with little snow can drop by **30 percent**.

⑥ Not all colour-changers work with fur. Some lizards, the cuttlefish, and many kinds of frog change their skin colour **within minutes**, not months. They have special skin cells called *chromatophores* that open and close like tiny coloured umbrellas, allowing the animal to blend with a leaf, a piece of bark, or even a patterned aquarium. The octopus is a champion of this: it can copy not only colour but also the **bumpy texture** of the coral it sits on.

⑦ Studying how and when animals change colour is not only interesting; it helps us **understand climate change**. If winter shortens faster than day length changes — and day length, as we know, cannot shorten at all — then more and more Arctic animals will find themselves wearing the wrong coat at the wrong time. The quiet white hare on a brown October field is, in a way, a small early warning.

⑧ So the next time you see a photograph of a snow-white fox crossing a field, remember: that fox is more than beautiful. It is a walking calendar, a pair of tiny hidden cells, and — increasingly — a fragile piece of our changing planet.

注释： hare 野兔；ermine 白鼬；ptarmigan 雷鸟；camouflage 伪装；pineal gland 松果腺；melatonin 褪黑激素；chromatophore 色素细胞

根据文章内容，选择最佳答案。

1. Which of the following animals does the article NOT list as an example of seasonal colour-changing?

A. Mountain hare
B. Arctic fox
C. Giant panda
D. Rock ptarmigan

2. What is the main reason animals change colour with the seasons?

A. To look pretty to tourists.
B. Camouflage — hiding from predators, or sneaking up on prey, by matching the colour of the surroundings.
C. To keep warm in winter.
D. To tell the time.

3. According to the article, what is the most important signal that tells a mountain hare to start turning white?

A. The temperature dropping below zero.
B. The first heavy snowfall of the year.
C. The shortening of daylight, which leads to more melatonin from the pineal gland and

less brown pigment in new hairs.
D. The call of other hares.

4. Why do scientists now worry about "mismatched hares"?

A. Because the hares are losing weight.
B. Because in warmer winters snow comes late, but day length has not changed, so hares turn white before the snow arrives and stand out against the brown field — their survival rates can drop by about 30%.
C. Because the hares eat too much.
D. Because the hares refuse to change colour.

5. What is special about octopuses compared with mountain hares?

A. Octopuses only change colour once a year.
B. Octopuses cannot change colour at all.
C. Octopuses change colour — and even the bumpy texture of their skin — in minutes, using special cells called chromatophores.
D. Octopuses change colour by eating snow.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段明列 mountain hare、Arctic fox、ermine、rock ptarmigan
【解析】大熊猫未被列入。A、B、D 均在例中。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "The reason is camouflage — the art of hiding in plain sight ... In nature, colour is survival."
【解析】季节换色是伪装求生。A、C、D 均非主要原因。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第④段 "The more important signal is day length, not temperature ... the pineal gland produces more of a chemical named melatonin ... tells the tiny cells that make fur pigment to stop producing brown colour."
【解析】日照变化触发松果腺→褪黑素→抑制色素。A、B、D 均不符。故选 C。

4.
【答案】B

【题型】推理判断题
【定位】第⑤段 "winters are becoming warmer ... hares still receive the old 'change now' signal ... survival rates in years with little snow can drop by 30 percent."
【解析】白毛 + 无雪 = 显眼 → 存活率下降。A、C、D 均不符。故选 B。

5.
【答案】C
【题型】细节理解题
【定位】第⑥段 "change their skin colour within minutes, not months ... chromatophores ... octopus is a champion ... copy not only colour but also the bumpy texture"
【解析】章鱼分钟级变色，甚至模仿质感。A、B、D 均错。故选 C。